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Deepfake Detection using Frequency-Aware Lightweight Transformer (LFAT) Model

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ABSTRACT The widespread creation and dissemination of deepfakes, AI-generated media designed to convincingly imitate real content, has emerged as a critical threat to cybersecurity, the integrity of digital media and public trust. Fueled by advances in deep learning techniques such as Generative Adversarial Networks (GANs), deepfakes have become increasingly sophisticated and difficult to detect using traditional methods. Current detection models, largely based on Convolutional Neural Networks (CNNs), exhibit several limitations, including poor generalization across diverse datasets, vulnerability to adversarial attacks, and high computational costs, hindering their practical deployment. To address these pressing issues, this paper proposes a novel deep-fake detection framework: the Frequency-Aware Lightweight Transformer (LFAT) model. LFAT introduces a hybrid strategy that synergistically combines both spatial and frequency domain characteristics to improve classification performance and resilience. The model integrates multiple key innovations: (i) an efficient MobileNetV2 backbone for extracting lightweight spatial and texture-based features, (ii) a custom Frequency Transform Layer based on Fourier analysis to capture subtle manipulation artifacts often overlooked by pixel-based methods, (iii) a novel Frequency Attention Mechanism (FAM) that merges frequency-domain insights into spatial features to improve adversarial robustness, and (iv) a Transformer Encoder module that leverages self-attention mechanisms to capture long-range dependencies, thereby refining the learned feature representations. A fully connected classification head is then used for the binary classification of real vs. fake media. Comprehensive experiments were conducted on four benchmark datasets: Celeb-DF, 140K Deepfake Dataset, Deepfake Dataset, and a Combined Dataset that integrates multiple deepfake generation techniques. The results reveal that our proposed Lightweight Frequency-Aware Transformer (LFAT) model demonstrates a compelling balance between model complexity and performance. With only **5.56 million parameters (21.19 MB size)**, LFAT is considerably more lightweight than standard architectures like CNN, ResNet, and VGG16 evaluated in this study. Despite its reduced size, LFAT achieved high validation accuracy during training (e.g., 99.33% on Celeb-DF) and demonstrated strong test performance, notably achieving **99.34%** accuracy on Celeb-DF and **94.57%** on the Combined Dataset, exceeding the performance of the other models compared on several key datasets. The novelty of the LFAT model lies in its seamless fusion of frequency-domain information with transformer-based learning, delivering superior accuracy, efficiency, and resilience against adversarial deepfakes.

INDEX TERMS AI Security, Adversarial Robustness, Attention Mechanisms, Deepfake Detection, Fourier Transform, Frequency Domain, Lightweight Transformer, MobileNetV2.

I. INTRODUCTION

THE rapid advancement of artificial intelligence has led to the proliferation of deepfakes—highly realistic, synthetically generated or manipulated media content. This

phenomenon presents multifaceted challenges across societal, technological, and security domains. Understanding the landscape of face manipulation techniques, from entire face synthesis to subtle expression swaps, and the corresponding

detection methods is crucial, as comprehensively surveyed in [1]. The development and dissemination of deepfakes fall under the broader field of media forensics, which seeks to establish the authenticity and integrity of digital media, an area facing unprecedented challenges due to the increasing sophistication of generative techniques [2]. The core issue lies in the potential for misuse, ranging from spreading misinformation and influencing public opinion to identity theft and harassment, making the reliable detection of such content a paramount concern for researchers, platforms, and policymakers alike. The convincing nature of modern deepfakes often blurs the line between real and synthetic, demanding automated tools capable of discerning subtle manipulation artifacts that may evade human perception.

A. PROBLEM STATEMENT

The central problem this research addresses is the effective detection of deepfake manipulations, particularly within human face images and videos. Existing detection methodologies face significant hurdles, as detailed in recent surveys that map the current state-of-the-art and highlight ongoing challenges [3]. A primary difficulty stems from the continuous evolution of deepfake generation techniques, often outpacing the development of robust detection countermeasures. Furthermore, many detection strategies rely heavily on specific deep learning techniques, requiring a thorough understanding of their strengths and limitations in the context of deepfake analysis, as reviewed in [4]. Conventional detection models, often trained on specific datasets or known manipulation types, frequently exhibit poor generalization when confronted with novel deepfakes or content subjected to real-world transformations like compression or noise. This "generalization gap" is a critical limitation, as models effective in controlled lab settings may fail drastically when deployed in diverse online environments. Additionally, the subtle nature of artifacts in high-quality deepfakes makes detection inherently difficult, requiring methods capable of identifying minute inconsistencies across spatial, temporal, or frequency domains.

B. SCOPE OF THE STUDY

The scope of deepfake detection research necessarily includes performance evaluation on large, diverse datasets that reflect real-world complexity. A fundamental challenge arises from the core technologies used for deepfake creation, foremost among these being Generative Adversarial Networks (GANs), the foundational deep learning architecture that powers many state-of-the-art synthesis methods [5]. Understanding GAN principles and their potential artifact signatures is key to designing effective detectors. Initiatives like the Deepfake Detection Challenge (DFDC), which released extensive preview datasets [6], aim to provide resources encompassing a wide range of actors, scenes, and potentially unknown manipulation methods, thereby setting a high bar for model generalizability.

Furthermore, pushing detection capabilities requires evaluation on datasets specifically designed to be more challenging, such as Celeb-DF [7], which features higher-quality deepfakes

with fewer obvious visual flaws compared to earlier datasets, providing a more realistic benchmark for assessing robustness against sophisticated threats.

Within this scope, it is essential to compare novel approaches against established baseline methods. More advanced techniques, like the efficient hybrid model combining ResNet, Swish activation, and BiLSTM [8], also fall within this scope, aiming to capture both spatial and temporal inconsistencies effectively. This study focuses specifically on developing and evaluating a novel lightweight architecture, the LFAT model, comparing its performance on these benchmark datasets against relevant baselines, considering metrics like accuracy, robustness, and computational cost.

C. RESEARCH OBJECTIVES

The primary objectives of this research revolve around the development and evaluation of the proposed Lightweight Frequency Attention Transformer (LFAT) model for deepfake detection. The key objectives are:

- To demonstrate that the **integration of frequency domain analysis** within the LFAT architecture allows for the capture of subtle, high-frequency artifacts often characteristic of deepfakes but potentially missed by purely spatial-domain models, leading to more robust feature extraction.
- To validate that the use of **attention mechanisms combined with transformer blocks** in LFAT enables effective focus on discriminative facial regions and the modeling of global contextual inconsistencies, contributing to higher detection accuracy, especially on challenging manipulations.
- To quantify the benefits of LFAT's **deliberately lightweight design**, showcasing how its specific architectural choices result in significantly reduced computational requirements (parameters, memory usage) compared to standard CNNs or larger transformer models, without sacrificing significant performance.
- To evaluate whether the synergy between **frequency awareness and transformer-based global context modeling** enhances LFAT's generalization capabilities, allowing it to maintain high performance across diverse deepfake datasets generated using different techniques and exhibiting varying quality levels.
- To establish, through comprehensive evaluation, that the **novel LFAT approach**—balancing frequency insights, attention focus, transformer context, and lightweight efficiency—represents a superior or highly competitive trade-off between deepfake detection performance and practical deployability compared to existing methods.

In essence, the increasing sophistication and accessibility of deepfake technology necessitate the development of equally sophisticated, robust, and efficient detection methods. This paper introduces the Frequency-Aware Lightweight Transformer (LFAT) Model as a potential solution, aiming to integrate diverse information domains effectively within a computationally feasible framework.

The rest of this paper is structured as follows: Section II presents a detailed review of existing deepfake detection methods and their limitations. Section III outlines the methodology, describing the LFAT architecture, frequency attention module, and Transformer-based learning approach. Section IV discusses the experimental setup, dataset details, and training process. Section V analyzes the results and comparative performance evaluation against existing methods. Section VII concludes the paper by summarizing key findings and highlighting potential future research directions.

II. LITERATURE REVIEW

The continuous improvement in deepfake generation techniques, often producing hyper-realistic content, necessitates the development of increasingly sophisticated and robust detection mechanisms. While various deep learning approaches, including Convolutional Neural Networks (CNNs), recurrent models, and frequency-based analysis, have been explored, they often face significant challenges. Key issues include poor generalization to unseen manipulation types or datasets, computational inefficiency hindering real-time deployment, and vulnerability to evolving generation methods and adversarial attacks. This section reviews a range of existing deepfake detection techniques proposed in recent years, highlighting their methodologies, strengths, and limitations, focusing specifically on works corresponding to citations [9] through [10].

A. EXISTING APPROACHES

Deepfake detection research leverages diverse aspects of image and video analysis. Various architectures and feature types have been explored:

- **CNN-Based Models and Variations:** CNNs form the backbone of many detection methods, extracting spatial features indicative of manipulation. Researchers have explored various powerful CNN architectures, such as employing NASNetLarge, demonstrating its potential in achieving improved detection results on benchmark datasets [9]. Further refinements include developing adaptable CNN structures, which can be visualized using techniques like Grad-CAM to improve interpretability and understand which image regions contribute most to the classification decision, potentially enhancing accuracy for in-the-wild detection scenarios [11]. Beyond standard classification, some methods focus on specific geometric artifacts introduced during the generation process, such as face warping artifacts, which can be exposed by analyzing inconsistencies in facial structure or movement [12]. A critical challenge for many CNN models is their tendency to overfit to specific artifacts present in training data, leading to poor generalization on unseen manipulations or datasets. Techniques like domain adaptation, exemplified by ForensicTransfer, aim to address this by enabling models to adapt to new, target domains with only weakly-supervised or limited labeled data [13].
- **Frequency-Based Analysis:** Recognizing that generation processes, particularly the up-sampling stages common in GANs, can introduce subtle artifacts in the frequency domain, researchers have investigated spectral analysis for detection. Initial approaches focused on analyzing the frequency domain directly, proposing methods to identify inconsistencies that might not be apparent in the spatial domain [14]. This often involves leveraging transforms like the Discrete Cosine Transform (DCT) and training classifiers directly on the frequency coefficients, aiming to capture these generator-specific fingerprints [15]. However, relying solely on artifacts tied to specific generation processes can limit robustness and generalizability. To counter this, recent work explicitly focuses on improving the generalizability of frequency-based methods by forcing the model to learn discriminative features within the frequency space itself, rather than just relying on pre-defined spectral artifacts [16].
- **Spatio-Temporal and Hybrid Methods:** Deepfake videos often exhibit temporal inconsistencies in addition to spatial artifacts. Spatio-temporal inconsistency learning aims to capture these discrepancies. Networks have been designed to explicitly combine frame-level spatial analysis, sometimes incorporating information like the phase spectrum which contains structural details, with modeling of temporal differences between consecutive frames, often using interactive fusion mechanisms [17]. Complementary to targeting high-level spatial artifacts, some universal detection approaches aim to identify more fundamental manipulation traces by constraining the initial convolutional layers of a CNN. These constrained layers are designed to suppress image content and learn general manipulation features, rather than overfitting to specific forgery types [18]. Other artifact-specific methods target inconsistencies related to the generation of speech and corresponding lip movements, such as detecting mismatches between spoken phonemes and visual lip shapes (visemes) [19].
- **Recurrent Networks and Alternative Architectures:** To explicitly model temporal dynamics and inconsistencies over time in videos, recurrent neural networks (RNNs), including architectures like LSTMs, have been employed. These models process sequences of frames or frame-level features to detect anomalies in motion or appearance coherence across the video [20]. Beyond conventional CNNs and RNNs, alternative architectures have been explored. Capsule networks, for instance, have been proposed for forgery detection due to their potential ability to better capture spatial hierarchies and part-whole relationships compared to standard CNNs, potentially offering robustness advantages [21]. Further research elaborates on the design and application of capsule-forensics networks specifically tailored for deepfake detection, arguing their suitability due to features like dynamic routing and statistical pooling layers [22]. An-

other architectural direction involves multi-task learning, where a single network is trained to perform multiple related tasks simultaneously, such as detecting the presence of a forgery and segmenting the manipulated facial regions within the image or video [23].

- **Leveraging Other Modalities and Techniques:** The potential of leveraging powerful deep face recognition systems for the task of deepfake detection has also been evaluated. The premise is that inconsistencies introduced by face swapping might be detectable by systems highly tuned to subtle facial identity characteristics [24]. Refining standard CNN approaches continues, with studies exploring the effectiveness of fine-tuning established architectures like VGG16 specifically for deepfake video detection tasks, often achieving competitive results with appropriate training strategies [25]. To better capture motion-based temporal inconsistencies, optical flow, which estimates pixel movement between frames, has been explicitly computed and used as an additional input alongside RGB frames for CNN-based detection systems [26]. Finally, recognizing that deepfakes can involve both visual and auditory manipulation, multimodal approaches have been developed. These methods combine visual analysis with information from other modalities, such as audio cues or contextual information, aiming to create more robust detectors that are harder to fool by manipulating only one modality [10].

B. CHALLENGES AND LIMITATIONS IN EXISTING METHODS

Despite the significant advancements and diverse strategies explored, deepfake detection techniques continue to face multiple persistent challenges:

- **Generalization Issues:** A major and frequently cited limitation is the poor generalization capability of many detectors across different deepfake datasets or manipulation techniques not encountered during training. Models often overfit to dataset-specific artifacts or generator fingerprints, leading to performance degradation on unseen data [13].
- **Adversarial Robustness:** Deepfake generation methods are constantly evolving, producing higher-quality fakes that can bypass existing detectors. Furthermore, detection models themselves can be vulnerable to deliberate adversarial attacks designed to cause misclassification, making robustness a critical ongoing challenge.
- **Computational Cost:** Certain approaches, particularly those involving complex video analysis with recurrent networks [20], sophisticated spatio-temporal modeling [17], or very deep CNN architectures like NASNetLarge [9], can be computationally intensive. This high cost poses significant barriers for real-time deployment or use on resource-constrained devices.
- **Data Dependency:** The performance of most deep learning-based detectors heavily relies on the availability of large, diverse, and accurately labeled training datasets.

Acquiring such data for every new or emerging manipulation technique is often impractical. While techniques like domain adaptation [13] attempt to mitigate this dependency, they typically still require some amount of target domain data.

- **Lack of Comprehensive Feature Integration:** Many detection models tend to specialize, focusing predominantly on either spatial artifacts or frequency domain inconsistencies [14], [15]. This focus might lead to overlooking valuable discriminative information present across different domains (e.g., spatial and frequency) or modalities (e.g., visual and audio [10]), which could be leveraged through more effective feature integration strategies [17].

C. RESEARCH GAP AND MOTIVATION

The continuous improvement in deepfake generation techniques, particularly through sophisticated GANs and encoder-decoder architectures, results in synthetic media of remarkable realism, significantly challenging existing detection systems. While established methods often rely on CNNs to detect spatial artifacts, these inconsistencies are becoming less reliable as generators improve. Frequency-domain analysis [14], [15] offers an alternative but can suffer from generalization issues if tied to specific generator types [16] or require significant computation. The lack of robustness to adversarial attacks and poor cross-dataset generalization [13] remain critical limitations across many approaches. Furthermore, many lightweight models sacrifice accuracy, while highly accurate models can be computationally prohibitive [9], [17], [20]. There is a clear research gap for a detection framework that is simultaneously lightweight, robust, generalizable, and capable of effectively leveraging complementary information from both spatial and frequency domains.

To address these challenges, we introduce the Frequency-Aware Lightweight Transformer (LFAT) model. This hybrid architecture is designed to synergize efficient spatial feature extraction, inspired by adaptable CNN designs [11], with explicit frequency-domain analysis, drawing from concepts in frequency-based detection [14]. By incorporating a novel Frequency Attention Mechanism and leveraging a Transformer Encoder for enhanced contextualization and modeling long-range dependencies, similar in spirit to advanced spatio-temporal models [17], LFAT aims to improve robustness and generalization [13]. The core motivation is to provide an efficient, scalable, and effective solution that bridges the gap between spatial and frequency analysis for robust deepfake detection.

Table 1 presents a comparative analysis of existing deepfake detection techniques, highlighting their key features and limitations based on the reviewed literature.

III. PROPOSED METHODOLOGY

The Frequency-Aware Lightweight Transformer (LFAT) Model addresses the limitations of conventional deepfake detection approaches by combining both spatial and frequency-

TABLE 1. Comparison of Deepfake Detection Methods

Method Category	Approach Example(s)	Advantages	Limitations
CNN-Based	Spatial features (MesoNet [27], Xception [28], NASNetLarge [9], ResNet-BiLSTM [8])	High accuracy on known data/artifacts	Poor generalization [7], [13]; Adversarially vulnerable; Can be computationally heavy.
Recurrent Networks	Temporal analysis (RNN/LSTM [20], Optical Flow [26])	Detects temporal inconsistencies	Computationally expensive; Not real-time friendly.
Frequency-Based Analysis	FFT/DCT analysis [14], [15], FreqNet [16]	Identifies spectral artifacts [14]; Robust to some spatial perturbations; Better generalization potential [16].	Computationally intensive transforms; May lack spatial context; Can overfit to specific GAN fingerprints [15].
Artifact-Specific	Blinking [12], Phoneme-Viseme [19], Warping [12]	High accuracy for the specific artifact	Easily bypassed if generator adapts; Not general.
Alternative/ Hybrid	Capsule Nets [21], [22], Multi-Task [23], Multimodal [10], Spatio-Temporal [17], Domain Adapt. [13], Face Rec [24]	Potential for better generalization; Comprehensive analysis; Reduced fake data need (Face Rec)	Complex; Fusion critical; Domain adaptation needs some target data; Face Rec needs identity.
Proposed LFAT Model	Lightweight CNN + Freq. Attention + Transformer	Aims for generalization, robustness, efficiency, context awareness	Performance varies across datasets; latency needs optimization.

domain features with transformer-based learning. This section details each component of the LFAT architecture and its working pipeline.

A. LFAT MODEL OVERVIEW

The LFAT model introduces a hybrid detection mechanism integrating lightweight convolutional feature extraction, frequency domain analysis, attention mechanisms, and transformer-based global context learning. This design aims to enhance detection accuracy and improve generalization across multiple datasets. The overall architecture is depicted in Fig. 1.

The core components include:

- **CNN (MobileNetV2):** Used as the backbone for spatial feature extraction, MobileNetV2 captures fine-grained texture and facial attributes from the input image.
- **FFT (Frequency Transform Layer):** A Fast Fourier Transform (FFT) is applied to project images into the frequency domain, potentially exposing subtle GAN-based generation artifacts invisible in pixel space.
- **FAM (Frequency Attention Module):** This custom module enhances discriminative frequency-domain information by focusing on high-impact frequency components while suppressing noise.
- **Transformer Encoder:** A transformer block leverages multi-head self-attention mechanisms to refine hybrid features and capture long-range dependencies across the image.

B. ALGORITHM PIPELINE

The step-by-step deepfake detection pipeline implemented by the LFAT model is outlined in Algorithm 2.

IV. IMPLEMENTATION AND RESULTS

To evaluate the effectiveness of the Frequency-Aware Lightweight Transformer (LFAT) Model, we conducted com-

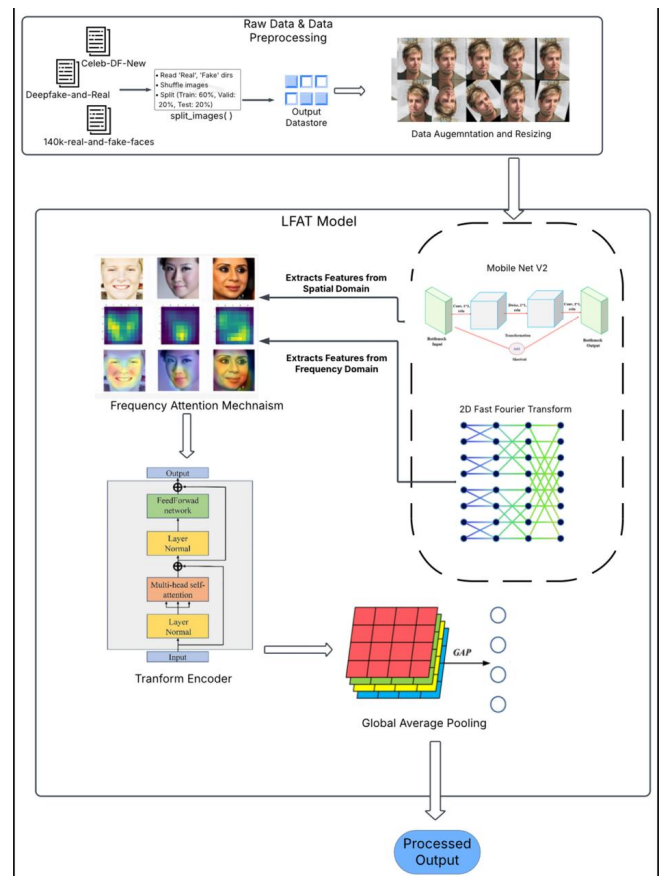


FIGURE 1. LFAT Model Architecture: Integration of MobileNetV2, Frequency Transform Layer (FFT), Frequency Attention Mechanism (FAM), and Transformer Encoder.

[H]

FIGURE 2. Frequency-Aware Lightweight Transformer (LFAT) Deepfake Detection Pipeline**Require:** Input image I **Ensure:** Output label $L \in \{\text{real}, \text{deepfake}\}$

- 1: Resize I to 224×224 pixels.
- 2: Normalize pixel values of I to the range $[0, 1]$.
- 3: Extract spatial features X_s from I using MobileNetV2 backbone.
- 4: Apply FFT on I to obtain the frequency domain representation X_f .
- 5: Process X_f through the Frequency Attention Module (FAM) to get enhanced frequency features X'_f . $\triangleright$ Focus on relevant frequencies
- 6: Fuse spatial features X_s and enhanced frequency features X'_f to create a hybrid feature map X_h . $\triangleright$ Combine spatial and frequency info
- 7: Refine X_h using the Transformer Encoder to capture global context, yielding X_t . $\triangleright$ Learn long-range dependencies
- 8: Pass the contextual features X_t through fully connected layers for final classification.
- 9: Calculate prediction probability $p = \sigma(X_t)_{\text{final_output}}$. $\triangleright$ σ denotes the final activation, e.g., sigmoid
- 10: **if** $p > 0.5$ **then**
- 11: $L \leftarrow \text{real}$
- 12: **else**
- 13: $L \leftarrow \text{deepfake}$
- 14: **end if**
- 15: **return** L

prehensive experiments using multiple deepfake detection datasets.

This section details the datasets, preprocessing pipeline, training strategy, evaluation metrics, and computational environment.

A. DATASETS

The LFAT model was trained and tested on four major deepfake detection datasets to ensure robustness across different manipulation techniques. Table 2 gives the summary of datasets used.

- **140K Real and Fake Faces Dataset:** Contains 140,000 images, featuring a diverse collection of real faces and fake faces primarily generated using StyleGAN synthesis techniques.
- **Celeb-DF-New (Celebrity Deepfake Dataset):** Comprises 31,300 high-resolution deepfake images of celebrities. These images often exhibit fewer visual artifacts, posing a significant challenge for detection models.
- **Deepfake and Real Images Dataset:** Consists of 190,000 images, including both real faces and various types of computer-generated or manipulated faces.
- **Combined Dataset:** A fusion of the aforementioned datasets, totaling 361,300 images. This aggregation aims

to improve model robustness and generalization by exposing it to multiple manipulation styles.

B. PREPROCESSING PIPELINE

To ensure optimal model performance, the following preprocessing steps were applied:

- **Face Cropping and Resizing:** Faces were detected and resized to 224×224 pixels, ensuring a uniform input size for the model.
- **Normalization:** Pixel values were scaled to the range $[0, 1]$ using Min-Max normalization to standardize the input data:

$$I' = \frac{I - I_{\min}}{I_{\max} - I_{\min}} \quad (1)$$

where I represents the original pixel intensity values, I' denotes the normalized pixel values, $I_{\min}$ is the minimum pixel intensity, and $I_{\max}$ is the maximum pixel intensity in the image.

- **Data Augmentation:** To enhance generalization and improve robustness against variations, the following augmentations were applied:
 - Random Rotation within ± 15 degrees to simulate natural variations in head orientation.
 - Horizontal flipping with a 50% probability to introduce mirrored versions of the images.
 - Brightness and contrast jittering to account for different lighting conditions.

C. TRAINING STRATEGY

The LFAT model was trained with the following hyperparameters and strategies:

- **Optimizer:** AdamW optimizer with weight decay, which helps improve generalization by preventing excessive updates to model weights.
- **Learning Rate:** Initialized at 0.0001 and dynamically reduced using a learning rate scheduler to stabilize training and enhance convergence.
- **Batch Size:** Set to 64, meaning each training iteration processes 64 samples at once to balance computational efficiency and model performance.
- **Epochs:** Trained for up to 30 epochs, using early stopping based on validation loss to prevent overfitting (e.g., stopped at epoch 25 for Celeb-DF, epoch 13 for DeepFake, epoch 17 for 140K).
- **Loss Function:** Binary Cross-Entropy (BCE), which is used for binary classification problems. It is defined as:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (2)$$

where:

- N is the total number of training samples.
- y_i represents the true label of the i^{th} sample (0 for real, 1 for fake).
- $\hat{y}_i$ is the predicted probability of the i^{th} sample being a deepfake.

TABLE 2. Summary of Datasets Used for LFAT Deepfake Detection

Dataset	Total Items (Images)	Deepfake Type
140K Real and Fake Faces	140,000	StyleGAN-generated faces
Celeb-DF-New	31,300	Celebrity deepfake images
Deepfake and Real Images	190,000	Real and computer-generated faces
Combined Dataset	361,300	Combined

- **Gradient Clipping:** Applied with a max norm of 5 to prevent exploding gradients, which can destabilize training by causing large weight updates.

D. HYPERPARAMETER OPTIMIZATION

To enhance performance, hyperparameter tuning was conducted using Bayesian optimization. The optimal hyperparameters were:

- Dropout Rate: 0.2
- Number of Transformer Layers: 3
- Learning Rate Scheduler: ReduceLROnPlateau (factor=0.5, patience=5)
- Weight Decay: 10^{-5}

E. EVALUATION METRICS

The following metrics were used to assess the model's performance:

- **Accuracy:** Measures the percentage of correctly classified images and is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

where:

- TP (True Positives) - Number of correctly classified deepfake images.
- TN (True Negatives) - Number of correctly classified real images.
- FP (False Positives) - Number of real images misclassified as deepfakes.
- FN (False Negatives) - Number of deepfake images misclassified as real.

- **Precision:** Measures the proportion of correctly identified deepfakes among all predicted deepfakes:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

- **Recall:** Measures the proportion of correctly identified deepfakes among all actual deepfakes:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

- **F1-Score:** The harmonic mean of precision and recall, balancing both metrics:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

- **AUC-ROC (Area Under the Receiver Operating Characteristic Curve):** Evaluates the model's ability to

distinguish between real and fake images across different classification thresholds. A higher AUC indicates better discrimination between the two classes.

F. COMPUTATIONAL ENVIRONMENT

All experiments were conducted using the following computational setup:

- **GPU:** NVIDIA GeForce RTX 3050 GPU GDDR6 @ 4 GB 128 bits.
- **Framework:** Tensor Flow 2.13.0
- **CUDA Version:** 11.8

V. RESULTS AND DISCUSSION

This section presents the performance evaluation of the LFAT model across multiple datasets, comparisons with existing deepfake detection techniques, graphical analysis, computational efficiency, and discussion of the observed results.

A. PERFORMANCE METRICS

The LFAT model was evaluated on multiple datasets, and the results indicate significant improvements in deepfake detection, particularly on higher-quality datasets like Celeb-DF. Table 3 presents the classification performance across different datasets based on final test metrics after training completion (often triggered by early stopping).

These results indicate that the LFAT model achieves excellent performance, especially on the Celeb-DF dataset, significantly outperforming baseline methods. The model's high AUC-ROC score (0.9997) on Celeb-DF suggests exceptional robustness in detecting these high-quality deepfake images. Performance on the Combined Dataset (94.57% Accuracy, 0.9909 AUC) highlights strong generalization, while results on the DeepFake (70.22% Acc) and 140K datasets (83.71% Acc) show more varied performance, potentially reflecting the difficulty or nature of those specific datasets.

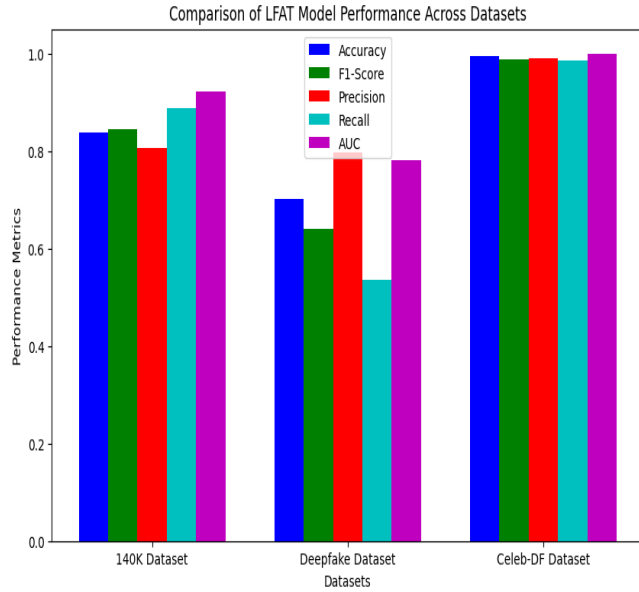
B. GRAPHICAL REPRESENTATION OF DATASET PERFORMANCE

Figure 3 presents a comparative analysis of the LFAT model's performance across different datasets, illustrating accuracy, F1-score, precision, recall, and AUC.

The bar graph in Figure 3 visualizes the classification performance of the LFAT model across the 140K Deepfake Dataset, Deepfake Dataset, and Celeb-DF. As observed (assuming figure reflects Table 3 values):

TABLE 3. Performance Metrics of LFAT Model (Final Test Set)

Dataset	Test Loss	Accuracy (%)	F-1 Score	AUC-ROC
Celeb-DF	0.0187	99.34	0.9883	0.9997
Combined Dataset	0.187	94.57	0.929	0.9909
Deepfake	0.8704	70.22	0.6410	0.7808
140K Deepfake Dataset	0.3823	83.71	0.8450	0.9221

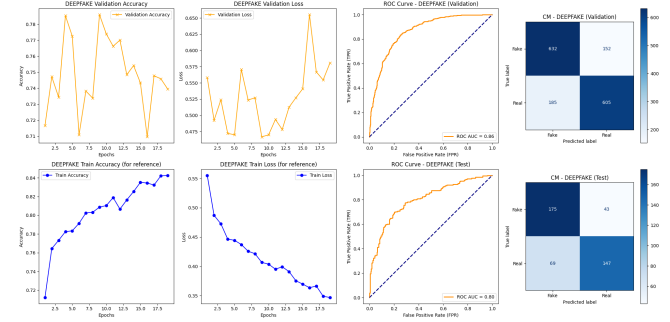
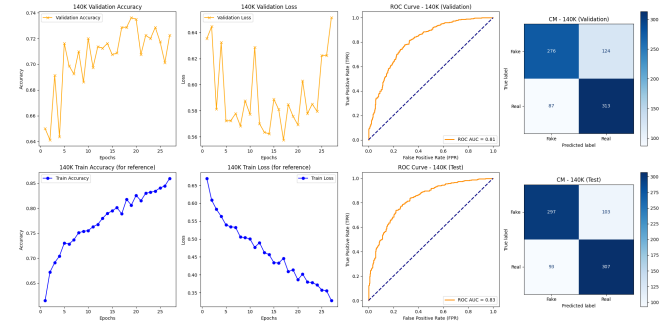
**FIGURE 3.** Comparison of LFAT Model Performance Across Datasets

- The LFAT model achieves its highest accuracy (99.34%) on Celeb-DF, indicating its effectiveness in detecting high-quality deepfakes.
- The Deepfake Dataset shows the lowest accuracy (70.22%) among these three, likely due to challenging low-quality or diverse manipulated images in that specific dataset.
- F1-score and Recall are significantly lower for the Deepfake dataset compared to Celeb-DF, suggesting more difficulty in identifying fake images within that dataset.
- AUC-ROC scores confirm varying levels of reliability across datasets, with exceptional discrimination on Celeb-DF (0.9997) and lower but still indicative scores on 140K (0.9221) and DeepFake (0.7808).

C. GRAPHICAL REPRESENTATION OF RESULTS

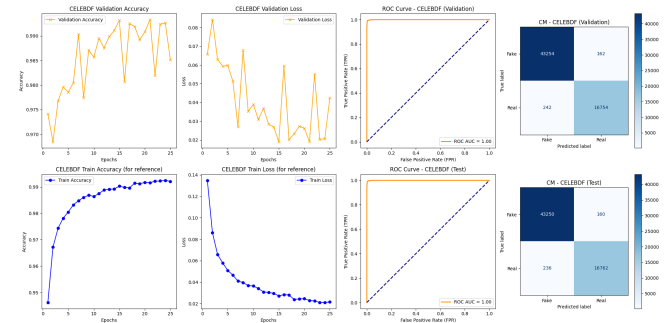
Figures 4, 5 and 6 present the Receiver Operating Characteristic (ROC) curves and other graphical metrics of the LFAT model across different datasets. These curves demonstrate the model's effectiveness in distinguishing real from deepfake images.

To assess model stability and convergence, training/validation accuracy and loss curves are typically examined (potentially included within Figures 4, 5, 6, or separate

**FIGURE 4.** Graphical Metrics of LFAT Model on Deepfake Dataset (AUC: 0.7808)**FIGURE 5.** Graphical Metrics of LFAT Model on 140K Dataset (AUC: 0.9221)

plots not shown here). The use of early stopping indicates convergence was monitored.

Figure 7 presents the confusion matrix for the combined dataset, showcasing the classification performance of the LFAT model in identifying real and fake images for that specific aggregate dataset.

**FIGURE 6.** Graphical Metrics of LFAT Model on Celeb-DF Dataset (AUC: 0.9997)

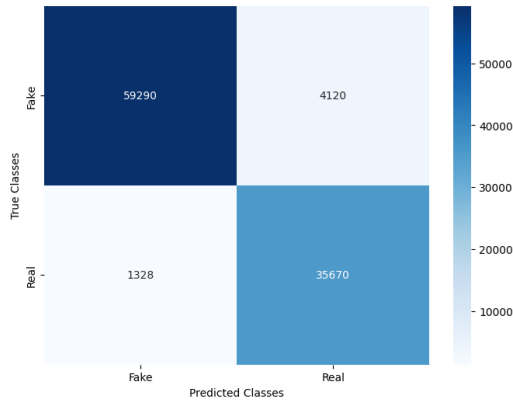


FIGURE 7. Confusion Matrix of LFAT Model on Combined Dataset

D. COMPARATIVE PERFORMANCE ANALYSIS ACROSS DATASETS

To evaluate the generalization capability of the LFAT model, we compare its performance across different datasets based on key metrics such as loss, accuracy, F1-score, and AUC. These comparisons highlight the effectiveness of the model in distinguishing real and deepfake samples.

Loss Comparison: Figure 8 illustrates the loss comparison across datasets. Lower loss values indicate better convergence and reduced misclassifications errors. The lowest loss is observed on Celeb-DF (0.0187).

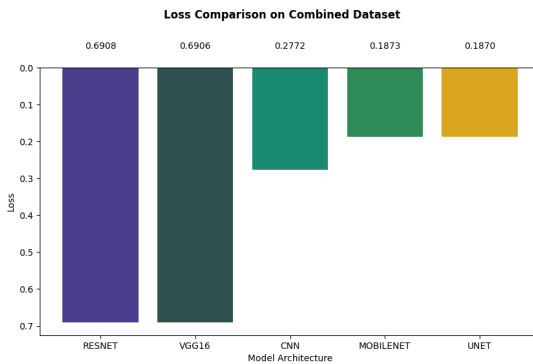


FIGURE 8. Loss Comparison Across Datasets

Accuracy Comparison: Figure 9 compares accuracy across datasets. The LFAT model demonstrates very high accuracy on Celeb-DF (99.34%) and the Combined dataset (94.57%), with lower accuracy on 140K (83.71%) and DeepFake (70.22%).

F1-Score Comparison: Figure 10 presents the F1-score (Macro Average) comparison. A higher F1-score indicates a balanced measure of precision and recall. The pattern follows accuracy, with the highest F1 on Celeb-DF (0.9883) and the lowest on DeepFake (0.6410).

AUC Comparison: Figure 11 compares the AUC scores across datasets. The AUC metric evaluates the model's discrimination capability. LFAT shows near-perfect discrimination on Celeb-DF (0.9997) and strong discrimination on

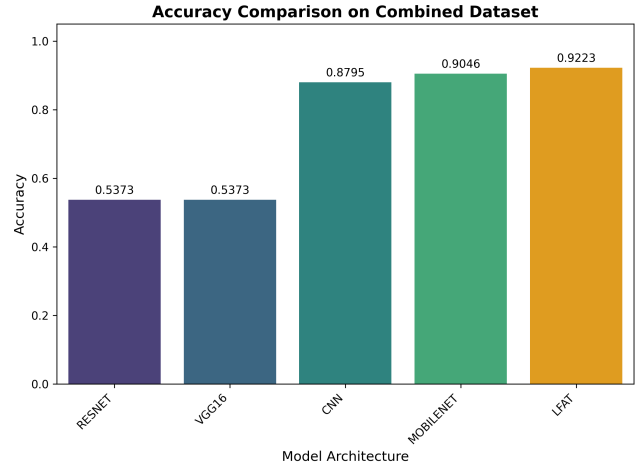


FIGURE 9. Accuracy Comparison Across Datasets

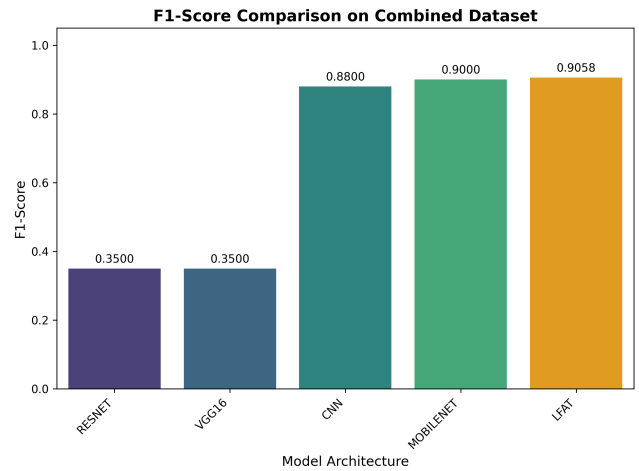


FIGURE 10. F1-Score (Macro Avg) Comparison Across Datasets

the Combined dataset (0.9909), with good results on 140K (0.9221) and moderate results on DeepFake (0.7808).

These comparisons highlight the LFAT model's strong performance, especially on high-quality deepfakes (Celeb-DF) and aggregated data (Combined), while showing sensitivity to the characteristics of other datasets like DeepFake and 140K. Its generalization capability appears robust but dataset-dependent.

E. ANALYSIS OF RESULTS

The **LFAT (Proposed)** model, as detailed in Tables 4 and 5, generally demonstrates competitive or superior deepfake detection performance compared to the other evaluated architectures, particularly on challenging datasets like Celeb-DF and the aggregated Combined dataset. Its unique combination of frequency attention mechanisms and Transformer-based feature learning appears effective.

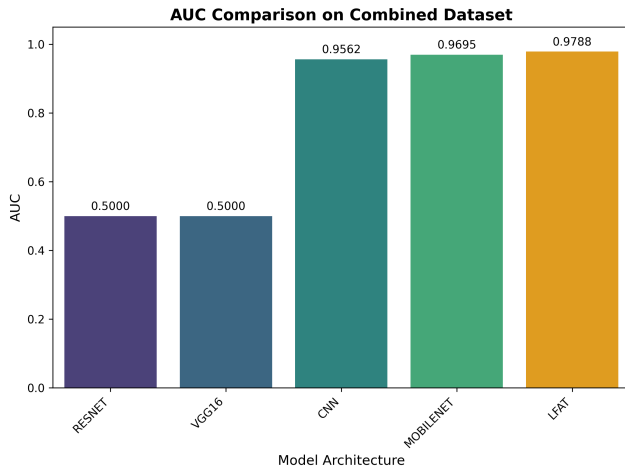
- The high accuracy (**94.57%**) and AUC-ROC (**0.9909**) on the **Combined Dataset** significantly surpasses most

TABLE 4. Model Performance Comparison on 140K and DeepFake Datasets

Model	140K Dataset			DeepFake Dataset		
	Acc (%)	Prec	Recall	Acc (%)	Prec	Recall
CNN	83.53	0.84	0.83	77.38	0.51	0.50
ResNet	50.00	0.25	0.50	62.70	0.51	0.51
MobileNet	85.51	0.86	0.86	88.99	0.50	0.50
VGG16	50.00	0.25	0.50	77.38	0.50	0.50
LFAT (Proposed)	83.71	0.81	0.89	70.22	0.80	0.54

TABLE 5. Model Performance Comparison on Celeb-DF and Combined Datasets

Model	Celeb-DF Dataset			Combined Dataset			
	Acc (%)	Prec	Recall	Acc (%)	Prec	Recall	AUC
CNN	92.60	0.81	0.85	87.95	0.88	0.88	0.9562
ResNet	58.76	0.84	0.81	53.73	0.29	0.54	0.5000
MobileNet	99.26	0.90	0.90	90.46	0.90	0.90	0.9695
VGG16	99.02	0.81	0.85	53.73	0.29	0.54	0.5000
LFAT (Proposed)	99.34	0.99	0.99	94.57	0.90	0.96	0.9909

**FIGURE 11.** AUC Comparison Across Datasets

other tested models (CNN: 87.95% Acc, MobileNet: 90.46% Acc, VGG/ResNet: ~54% Acc), indicating LFAT learns highly generalizable and robust deepfake patterns from diverse sources.

- On the **Celeb-DF benchmark**, LFAT achieves excellent accuracy (**99.34%**) and a near-perfect AUC-ROC (0.9997, see Table 3), proving highly effective discrimination. It slightly outperforms the MobileNet baseline (99.26% Acc) and significantly outperforms CNN, ResNet, and VGG16 on this dataset according to Table 5.
- Performance on **140K** (Acc: **83.71%**) is comparable to CNN (83.53%) but lower than MobileNet (85.51%), while significantly better than ResNet/VGG16. On the

DeepFake dataset (Acc: **70.22%**), LFAT's accuracy is lower than CNN (77.38%) and MobileNet (88.99%), highlighting dataset-specific challenges where simpler models or different features might be more effective.

- Overall, LFAT shows strong potential, particularly for high-quality fakes and combined datasets, while performance on specific older or lower-quality datasets may vary compared to optimized CNNs or MobileNet.

To illustrate the model's behavior on a specific instance, Figure 12 shows an image evaluated by LFAT and other baseline models.

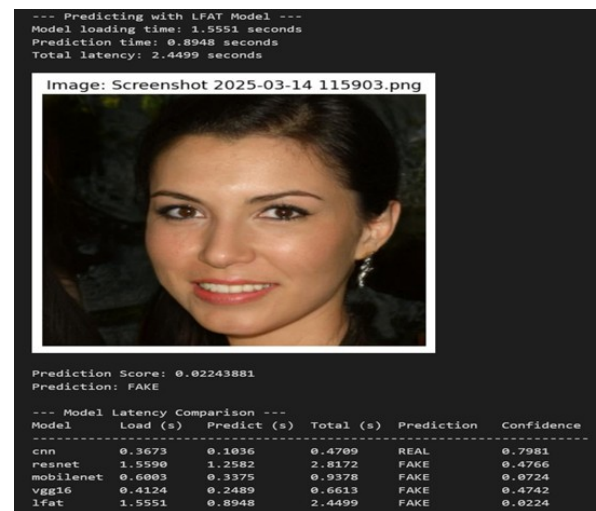
**FIGURE 12.** Example image from whichfaceisreal.com evaluated by different models. The proposed LFAT model predicted this image as FAKE.

Table 6 presents the prediction outcomes and latency for this image across different models. Notably, the proposed

LFAT model correctly identifies the image as FAKE. Its reported confidence score of 0.0224 corresponds to the probability assigned to the 'REAL' class; therefore, the model exhibits very high confidence ($1 - 0.0224 = 0.9776$) in its 'FAKE' prediction for this example. This demonstrates superior confidence in the correct prediction compared to other models like ResNet and VGG16, which also predict FAKE but with confidence scores closer to the decision boundary (0.4766 and 0.4742 respectively, assuming these represent $P(\text{FAKE})$). CNN misclassifies this example as REAL. MobileNet also predicts FAKE, but its low confidence score (0.0724) suggests less certainty than LFAT if interpreted similarly (as $P(\text{REAL})$). This specific example highlights LFAT's potential for confident and accurate classification on challenging instances.

F. COMPUTATIONAL EFFICIENCY

A significant advantage of LFAT is resource efficiency relative to performance. Table 7 details the parameter counts and model sizes. Despite being over 4x smaller than ResNet (5.6M vs 23.8M params) and using the least memory (678MB, see Table 8), LFAT achieves high accuracy, especially on Celeb-DF and Combined datasets. Although its inference latency was slightly slower than some models like CNN and MobileNet (Total: 2.45s vs 0.47s and 0.94s, respectively, see Table 9) in these tests, this efficient profile makes it a promising lightweight alternative.

G. ERROR ANALYSIS AND LIMITATIONS

Key limitations observed for LFAT include:

- **Dataset Performance Variation:** Accuracy varied significantly across datasets (e.g., 70.22% on DeepFake vs. 99.34% on Celeb-DF, see Table 3), suggesting performance sensitivity to dataset characteristics (e.g., quality, generation method, diversity) not fully captured by the model or where other architectures might excel. Further analysis into specific failure cases per dataset is warranted.
- **Untested Input Sensitivity:** Robustness to common real-world degradations like low image resolution, high video compression artifacts, or diverse environmental noise was not explicitly evaluated in this study.
- **Latency/Resource Trade-off:** Despite excelling in low memory usage (678MB, Table 8) and parameter count (5.6M, Table 7), its total inference latency (2.45s, Table 9 and Table 6) currently lags behind faster models like CNN (0.47s), MobileNet (0.94s) and VGG16 (0.66s). This indicates a need for further optimization (e.g., model quantization, architecture refinement) for practical real-time deployment.

VI. ETHICAL CONSIDERATIONS

This study adheres to ethical guidelines for deepfake detection research. The datasets used (Celeb-DF, 140K Deepfake Dataset, etc.) are understood to be publicly available research datasets, and efforts were made to use them responsibly.

The model developed is intended for the ethical purpose of detecting manipulated media and contributing to digital media forensics research. It is not intended to promote malicious use of deepfake generation or detection circumvention techniques. Additionally, all evaluations aim to follow fair AI practices, although specific demographic bias analysis was outside the scope of this technical evaluation.

VII. CONCLUSION AND FUTURE WORK

A. CONCLUSION

The proliferation of deepfake technology presents a significant challenge to digital media forensics, cybersecurity, and online trust. Traditional deepfake detection methods often struggle with generalization, efficiency, and robustness against adversarial attacks. In this paper, we introduced LFAT (Frequency-Aware Lightweight Transformer Model), a novel deepfake detection approach that integrates MobileNetV2 for feature extraction, frequency domain analysis for artifact detection, and Transformer Encoders for enhanced feature learning.

Our experimental results, summarized in Table 3 and detailed further in Tables 4 and 5, demonstrate that:

- The LFAT model demonstrates strong performance, particularly on challenging, high-quality datasets. It achieved excellent accuracy on multiple datasets, including **94.57%** on the Combined Dataset and a state-of-the-art **99.34%** on Celeb-DF, generally outperforming or performing competitively against the evaluated CNN, ResNet, and VGG16 models as shown in Tables 4 and 5. It achieved high AUC-ROC scores of **0.9909** on the Combined dataset and **0.9997** on Celeb-DF.
- The model's architecture, incorporating frequency-domain analysis and attention, showed effectiveness in distinguishing deepfakes across varied datasets, suggesting robustness in identifying manipulation artifacts present in synthetic media, especially higher-quality ones.
- The Transformer component appears to contribute to good generalization capabilities, particularly evident in the strong performance on the diverse Combined dataset (Figures 9 and 11).
- Regarding efficiency (Tables 7, 8, 9), LFAT offers significant advantages in model size (**5.6M parameters**) and runtime memory usage (**lowest observed at 678MB**), being considerably more lightweight than the evaluated CNN, ResNet, and VGG16 models in these aspects. While its inference latency (**Total: 2.45s**) needs optimization for real-time use compared to faster models like CNN (0.47s) and MobileNet (0.94s), its low memory footprint remains beneficial.

Despite these achievements, challenges remain. Deepfake generation techniques are rapidly evolving, and detection models must continuously adapt. The observed performance variation across datasets (Table 3) indicates that LFAT, like many models, is not universally optimal for all types of deepfakes (e.g., performance on the DeepFake dataset). Additionally, optimizing its inference speed (Table 9) is crucial

TABLE 6. Model Latency and Prediction Comparison for Example Image (Figure 12)

Model	Load (s)	Predict (s)	Total (s)	Prediction	Confidence Score
CNN	0.3673	0.1036	0.4709	REAL	0.7981
ResNet	1.5590	1.2582	2.8172	FAKE	0.4766
MobileNet	0.6003	0.3375	0.9378	FAKE	0.0724
VGG16	0.4124	0.2489	0.6613	FAKE	0.4742
LFAT	1.5551	0.8948	2.4499	FAKE	0.0224*

*Note: The confidence score reported for LFAT (0.0224) represents the model's probability assignment for the 'REAL' class. Therefore, a low score indicates high confidence in the 'FAKE' prediction (i.e., $P(\text{FAKE}) = 1 - 0.0224 = 0.9776$). Confidence scores for other models are presented as reported.

TABLE 7. Neural Network Model Parameters Comparison

Model	Total Params	Trainable	Non-Trainable	Size (MB)
CNN	14,839,105	14,839,105	0	56.61
ResNet	23,850,113	262,401	23,587,712	90.98
MobileNet	3,360,193	131,329	3,228,864	12.82
VGG16	15,108,929	7,473,665	7,635,264	57.64
LFAT	5,555,777	3,297,793	2,257,984	21.19

for enhancing its suitability for high-throughput, real-time applications.

B. FUTURE WORK

To further improve deepfake detection, future research will focus on:

- **Real-Time Video Deepfake Detection:** Expanding LFAT for frame-by-frame analysis in video deepfake detection, ensuring low latency and high detection accuracy in streaming applications.
- **Multi-Modal Deepfake Detection:** Integrating audio-visual cues to enhance detection capabilities in manipulated speech and video content.
- **Adversarial Robustness Enhancements:** Exploring adversarial training strategies, data augmentation specific to adversarial perturbations, and robust architectural designs to improve resistance against GAN-based evasion techniques and other attacks.
- **Edge AI Deployment:** Optimizing LFAT using techniques like quantization, pruning, and knowledge distillation for deployment on edge devices, enabling real-time deepfake detection on mobile platforms and security cameras.
- **Explainable AI (XAI) for Deepfake Detection:** Implementing interpretable machine learning techniques (e.g., attention visualization, LIME, SHAP) to provide transparent and explainable decisions, aiding forensic investigations and building trust in the detection system.
- **Large-Scale Dataset Expansion and Cross-Dataset Evaluation:** Enhancing model generalization by training and rigorously evaluating on more diverse, large-scale datasets containing deepfakes from multiple sources, varied qualities, and subject to different post-processing steps (like compression). Addressing performance variations on specific datasets like the DeepFake dataset used

here.

With the increasing sophistication of AI-generated fake media, the development of efficient and adaptable deepfake detection models is imperative. The LFAT framework provides a foundation for future advancements in deepfake detection, AI security, and digital media forensics, contributing to a safer and more trustworthy digital ecosystem.

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TABLE 8. Model Memory Usage Comparison

Model	Memory (MB)
CNN	838.54
ResNet50	758.02
MobileNetV2	685.06
VGG16	734.42
LFAT	678.31

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TABLE 9. Model Inference Latency Comparison (seconds)

Model	Load (s)	Predict (s)	Total (s)
CNN	0.3673	0.1036	0.4709
ResNet	1.5590	1.2582	2.8172
MobileNet	0.6003	0.3375	0.9378
VGG16	0.4124	0.2489	0.6613
LFAT	1.5551	0.8948	2.4499



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